Ryan Huang

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Technical Skills

Programming Languages: C/C++, Python, Bash, Java, MATLAB, R
Firmware & Embedded Protocols: FreeRTOS, Linux, JTAG, GDB, UART, I2C, SPI, TCP/IP, SNMP, ICMP
Development Tools: Git, Docker, Wireshark, Altium Designer, SolidWorks

Education

The University of British Columbia — Vancouver, BC
Bachelor of Applied Science (BASc) in Biomedical Engineering — *Expected Graduation: May 2029*
Relevant Courses: Control Systems, Circuits & EM, Biomedical Instrumentation, MEMS Devices, DSP & DIP

Work Experience

Corinex — Jan. 2026 - Aug. 2026
Firmware Developer Co-op — *Vancouver, BC*

- Investigated embedded firmware failures using JTAG/GDB and implemented 20+ bug fixes
- Programmed a flash utility tool using a JTAG peripheral with GDB scripts to write firmware images directly to SPI flash through memory-mapped registers
- Redesigned the shell logging system by implementing a lock-efficient static ring buffer, reducing mutex contention and improved concurrent logging performance by 90%
- Developed BusyBox shell scripts to monitor PHY speed, uptime, and LTE RSSI for embedded device validation
- Designed an optical port converter PCB in Altium Designer to interface embedded communication hardware
- Developed a Python automated OCPP validation framework that reduced manual testing time by 90%
- Supported deployment, validation, and compliance testing of broadband-over-power-line systems for smart grid and EV charging applications
- More Information

Project Experience

Automated Finger Stretching Device for Quadriplegia — Sep. 2025 - Sep. 2026
Electrical & Embedded Team Lead on UBC BEST - ENABLE — *Vancouver, BC*

- Led the electrical and embedded systems development of a multi-motor finger stretching device for a quadriplegic client, translating design specifications into control system design
- Designed the embedded architecture by programming an Arduino multi-motor control framework, assembling the electrical system, and reducing wiring complexity for scalable actuator control
- Co-authored a peer-reviewed research abstract published in the *Gerontechnology* Journal and presented the project at the International Society for Gerontechnology Conference
- Publication Link, Project Details

Embedded Music Controller

- Developed an Arduino music controller that directly accesses Spotify to display current song and artist along with volume and playback control
- Implemented bidirectional serial communication between Arduino firmware and a Python script with robust encoding and decoding, achieving perfect transmission reliability over 200+ test cycles
- Integrated the Spotify Web API using OAuth authentication to securely synchronize playback state between the controller and Spotify in real time
- More Projects & Highlights

Awards

UBC Elizabeth and Leslie Gould Entrance Scholarship for Engineering
Merit-based scholarship renewed annually for academic excellence, leadership, and co-curricular achievement in the BASc in Engineering program.

Hobbies: piano, music curation, basketball, ultimate frisbee, prototyping